GREEEN tallgrass prairie restoration: Seeded richness ~ Species pool size (Lux Arbor)

Markus Bauer

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Markus Bauer

Technichal University of Munich, TUM School of Life Sciences, Chair of Restoration Ecology, Emil-Ramann-Straße 6, 85354 Freising, Germany

markus1.bauer@tum.de

ORCiD ID: 0000-0001-5372-4174 Google Scholar GitHub: markus1bauer

> **NOTE:** To compare different models, you only have to change the models in the section 'Load models'

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```r
library(here)
library(tidyverse)
library(ggbeeswarm)
library(patchwork)
library(lme4)
library(glmmTMB)
library(DHARMa)
library(emmeans)
```

Packages

```r
sites <- read_csv(
  here("data", "processed", "data_processed_sites.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    id_plot = "f",
    year = "f",
    seeding_time = col_factor(
      levels = c("unseeded", "fall", "spring"), ordered = FALSE
      ),
    seeded_pool = col_factor(
      levels = c("0", "6", "12", "18", "33"), ordered = TRUE
      ),
    treatment_id = "f",
    richness_type = "f"
  )
) %>%
  filter(
    site == "Lux Arbor",
    richness_type == "seeded_richness",
    !(treatment_id %in% c("2", "4"))
  ) %>%
  mutate(y = richness_1qm + richness_25qm)
```

Load data

Statistics

Data exploration

Means and deviations

```r
Rmisc::CI(sites$y, ci = .95)
```

```
##    upper     mean    lower
## 8.693704 8.128920 7.564136
```

```r
median(sites$y)
```

```
## [1] 7
```

```r
sd(sites$y)
```

```
## [1] 4.861088
```

```r
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##   5%  95%
##  0.3 16.0
```

```r
sites %>% count(year)
```

```
## # A tibble: 6 x 2
##   year      n
##   <fct> <int>
## 1 2015     47
## 2 2016     48
## 3 2017     48
## 4 2018     48
## 5 2019     48
## 6 2025     48
```

```r
sites %>% count(seeded_pool)
```

```
## # A tibble: 4 x 2
##   seeded_pool     n
##   <ord>       <int>
## 1 6              72
## 2 12             72
## 3 18             72
## 4 33             71
```

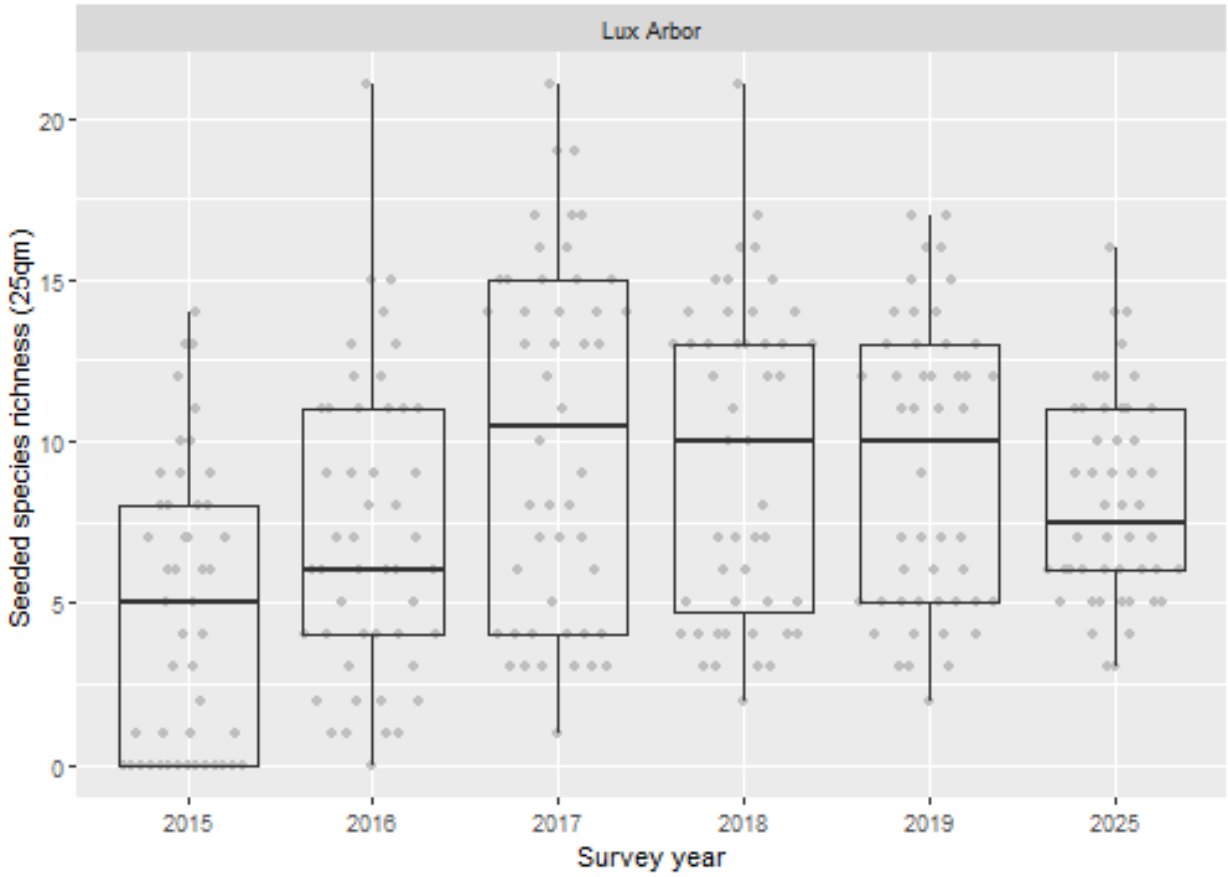
```r
sites %>% count(seeding_time)
```

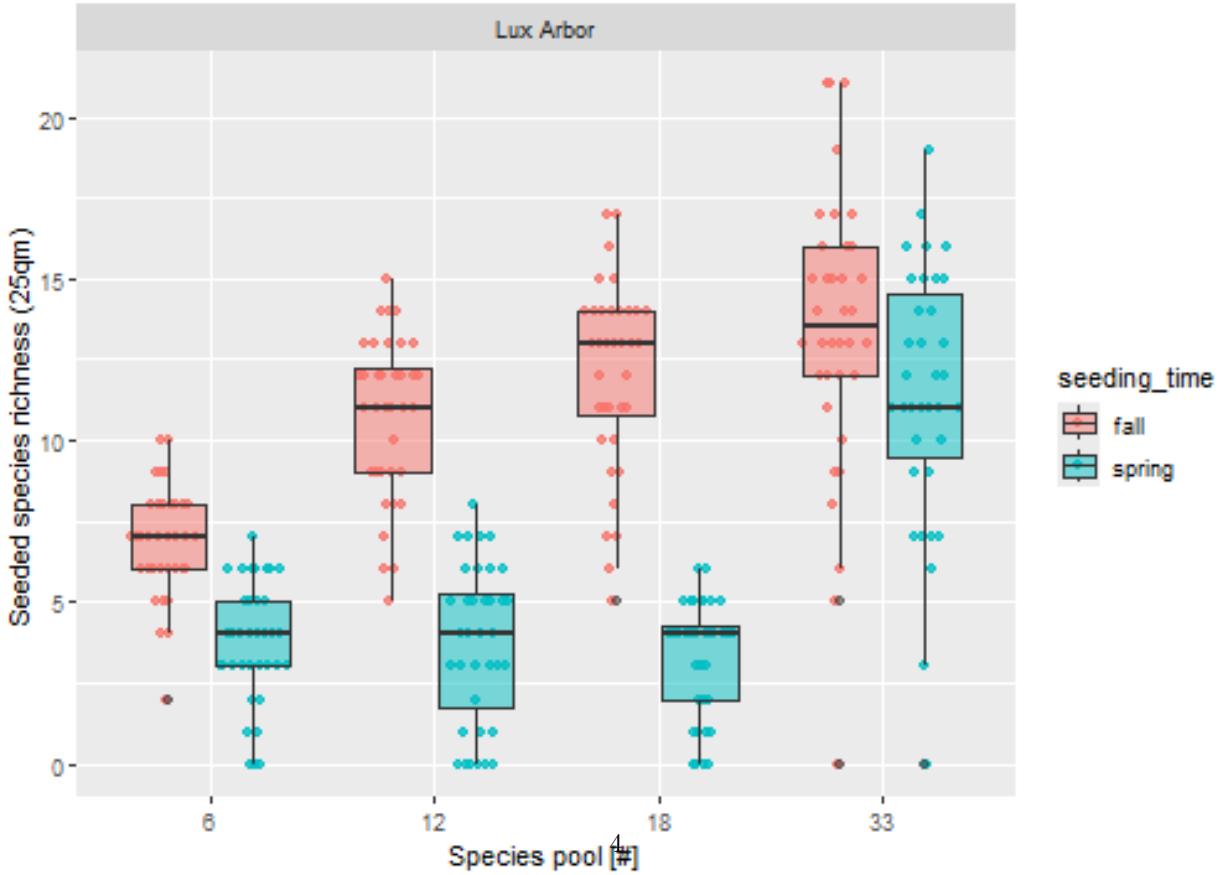
```
## # A tibble: 2 x 2
##   seeding_time     n
##   <fct>        <int>
## 1 fall           144
## 2 spring         143
```

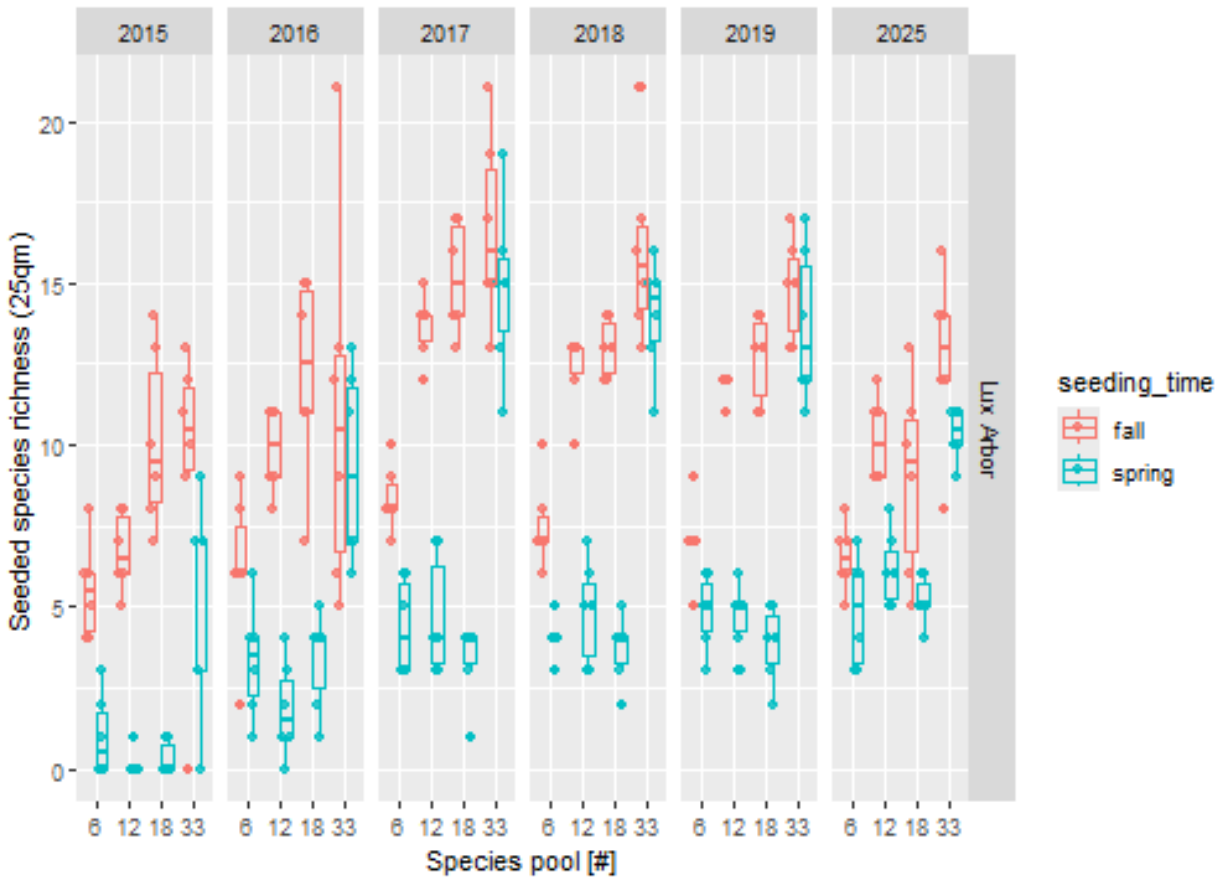
```r
sites %>% count(herbicide)
```

```
## # A tibble: 2 x 2
##   herbicide     n
##       <dbl> <int>
## 1         0   144
## 2         1   143
```

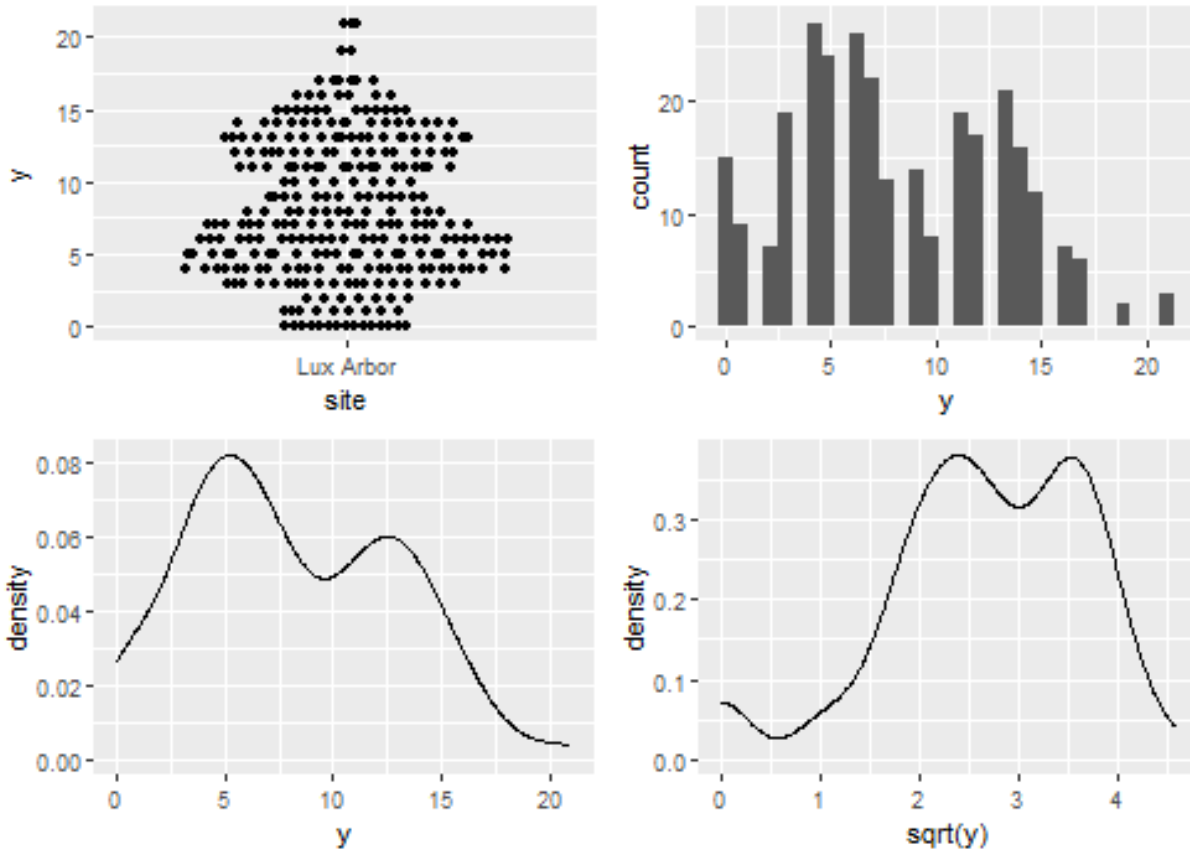
Graphs of raw data (Step 2, 6, 7)






2015
2016
2017
2018
2019
2025
Lux Arbor
Seeded species richness (25qm)
0
5
10
15
20
6 12 18 33
Species pool [#]
seeding_time
fall
spring

Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude r > 0.7 Dormann et al. 2013 Ecography DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cwm.")) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#     ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

```
load(file = here("outputs", "models", "model_richness_pool_lux_1.Rdata"))
load(file = here("outputs", "models", "model_richness_pool_lux_2.Rdata"))
m_1 <- m1
m_2 <- m2
```

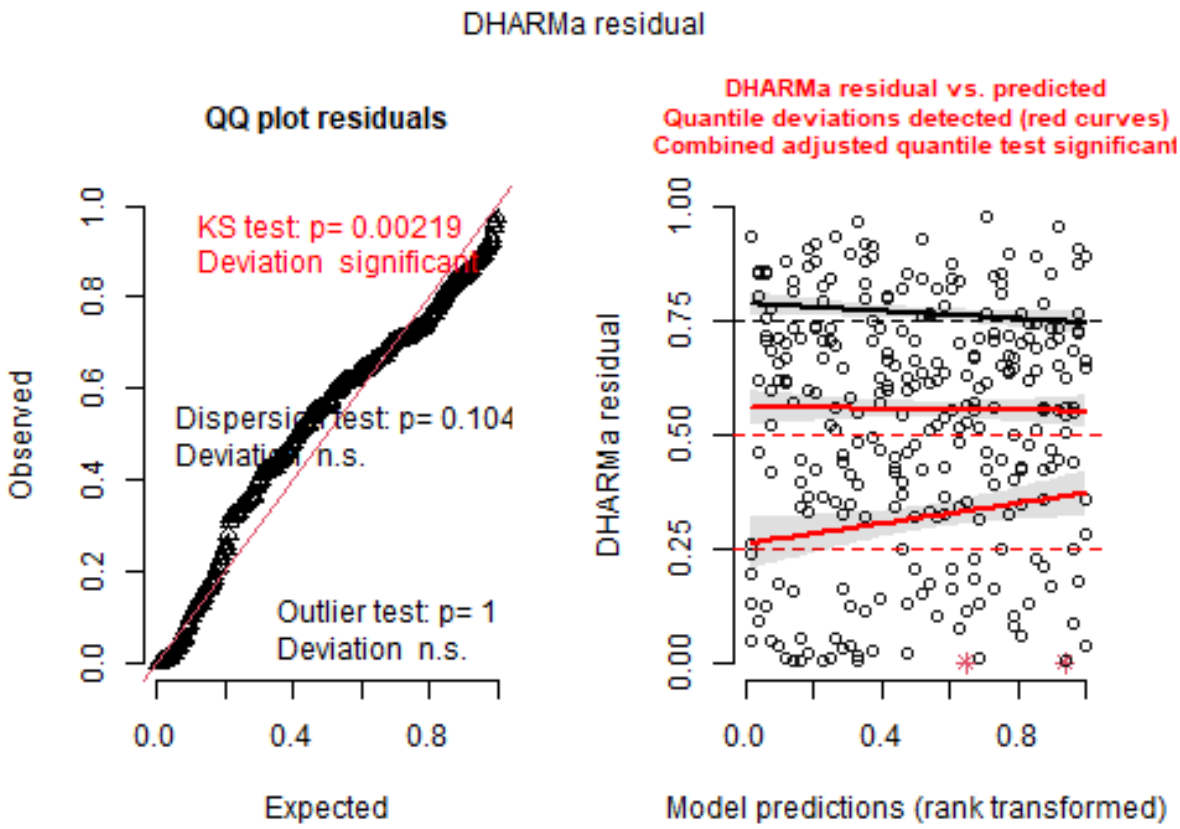
```
formula(m_1)
## y ~ seeded_pool * seeding_time + water_cap + (1 | year)
formula(m_2)
```

```
## sqrt(y) ~ seeded_pool * seeding_time * year + water_cap
```

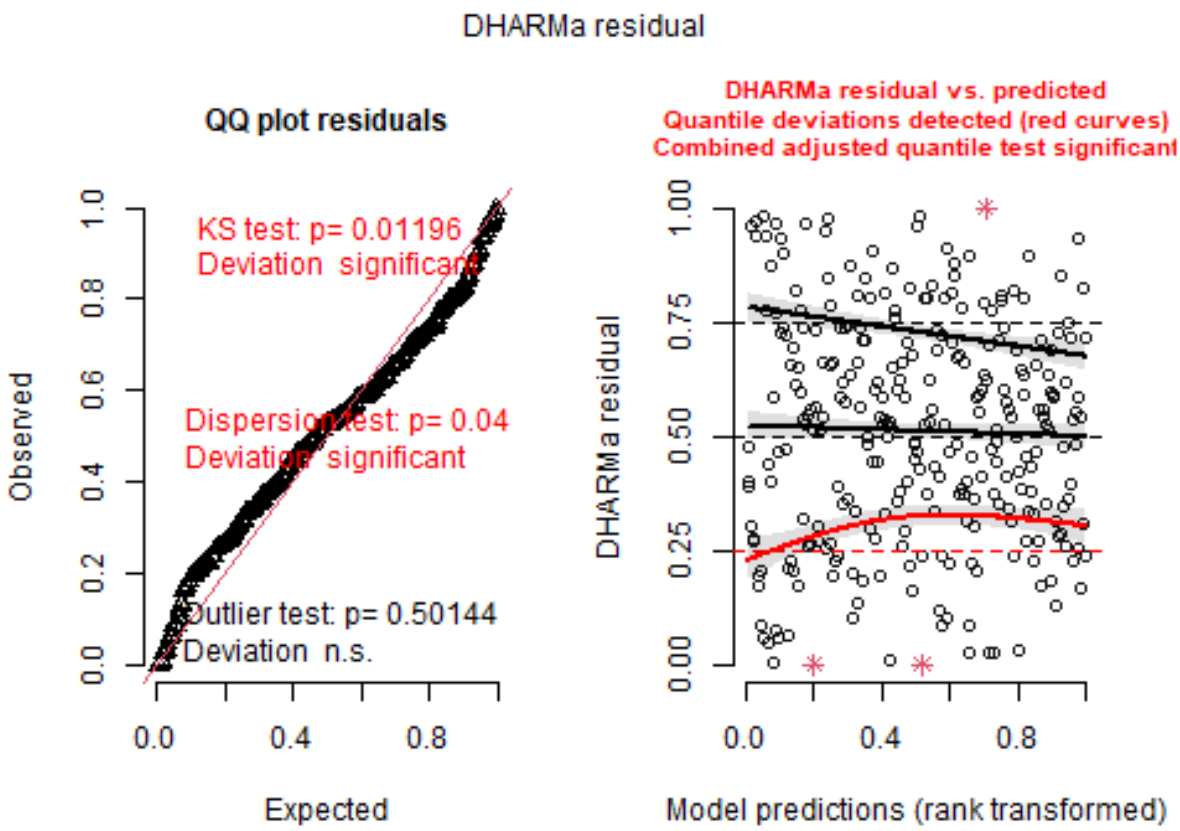
Model check

DHARMa

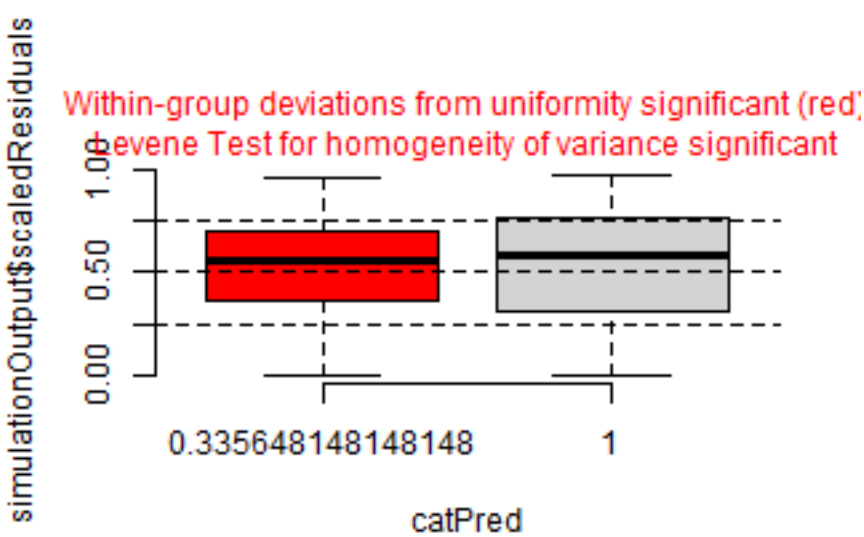
```
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```



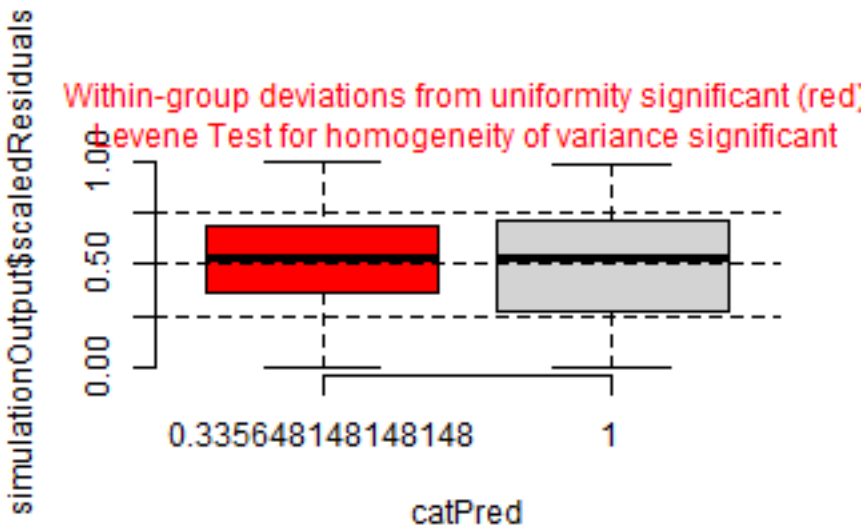

```
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```



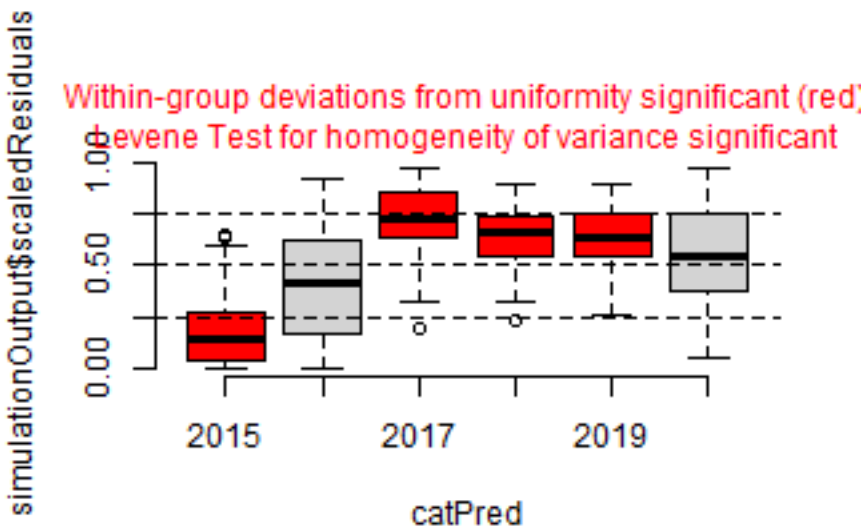

```
plotResiduals(simulation_output_1$scaledResiduals, sites$herbicide)
```



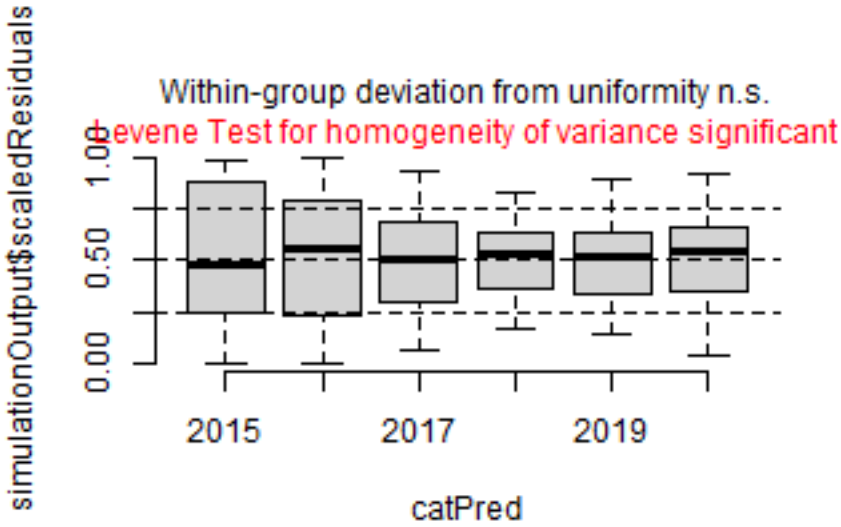

```
plotResiduals(simulation_output_2$scaledResiduals, sites$herbicide)
```



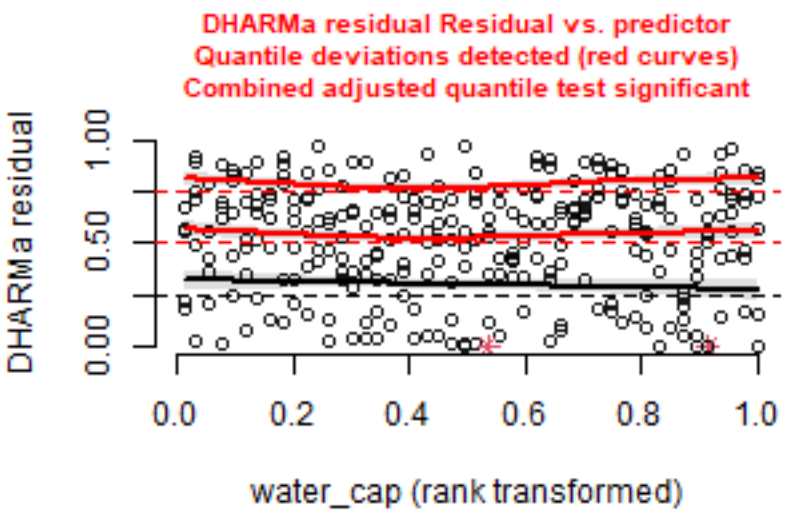

```
# plotResiduals(simulation_output_1$scaledResiduals, sites$seeding_time)
# plotResiduals(simulation_output_2$scaledResiduals, sites$seeding_time)
plotResiduals(simulation_output_1$scaledResiduals, sites$year)
```



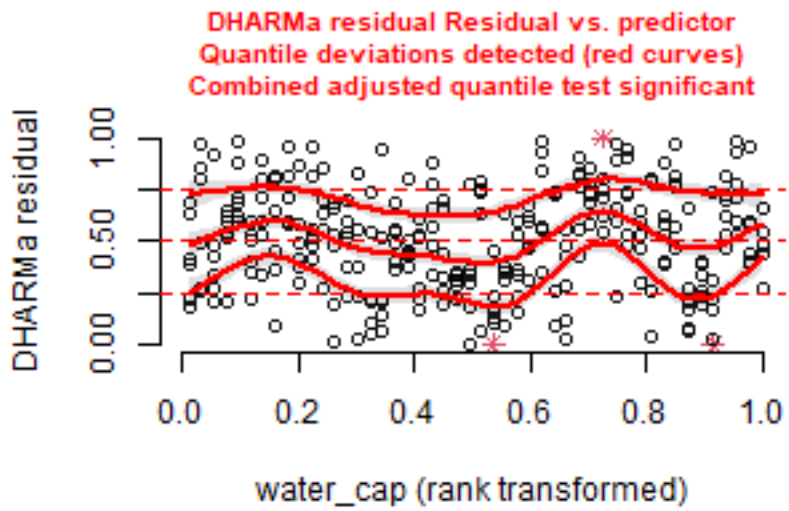

```
plotResiduals(simulation_output_2$scaledResiduals, sites$year)
```




```
plotResiduals(simulation_output_1$scaledResiduals, sites$water_cap)
```




```
plotResiduals(simulation_output_2$scaledResiduals, sites$water_cap)
## Warning in newton(lsp = lsp, X = G$X, y = G$y, Eb = G$Eb, UrS = G$UrS, L = G$L,
## : Anpassung beendet mit Schrittweitenfehler - Ergebnisse sorgfältig prüfen
## Warning in newton(lsp = lsp, X = G$X, y = G$y, Eb = G$Eb, UrS = G$UrS, L = G$L,
## : Anpassung beendet mit Schrittweitenfehler - Ergebnisse sorgfältig prüfen
```



Check collinearity part 2 (Step 5)

Remove VIF > 3 or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
# car::vif(m_1)
car::vif(m_2)
```

```
## there are higher-order terms (interactions) in this model
## consider setting type = 'predictor'; see ?vif
```

```
##                                       GVIF Df GVIF^(1/(2*Df))
## seeded_pool                   1.784315e+03  3        3.482667
## seeding_time                  6.130766e+00  1        2.476038
## year                          3.144251e+01  5        1.411730
## water_cap                     1.336977e+00  1        1.156277
## seeded_pool:seeding_time      1.888629e+03  3        3.515802
## seeded_pool:year              7.077888e+06 15        1.691726
## seeding_time:year             1.140502e+02  5        1.605867
## seeded_pool:seeding_time:year 7.520530e+06 15        1.695151
```

Model comparison

R2 values

```
MuMIn::r.squaredGLMM(m_1)
## Warning: the null model is only correct if all the variables it uses are identical
## to those used in fitting the original model.
##                  R2m       R2c
## delta      0.6353779 0.7518700
## lognormal  0.6444658 0.7626241
## trigamma   0.6254531 0.7401255
MuMIn::r.squaredGLMM(m_2)
##             R2m       R2c
## [1,] 0.8142087 0.8142087
```

AICc

Use AICc and not AIC since ratio n/K < 40 Burnahm & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
MuMIn::AICc(m_1, m_2) %>%
  arrange(AICc)
##     df      AICc
## m_2 50  418.2479
## m_1 11 1329.5797
```

Predicted values

Summary table

```
car::Anova(m_2, type = 3)
```

```
## Anova Table (Type III tests)
##
## Response: sqrt(y)
##                                Sum Sq  Df  F value    Pr(>F)
## (Intercept)                    25.871   1 130.3470 < 2.2e-16 ***
## seeded_pool                     2.266   3   3.8052   0.01082 *
## seeding_time                   42.765   1 215.4676 < 2.2e-16 ***
## year                           13.586   5  13.6906 9.421e-12 ***
## water_cap                       0.808   1   4.0729   0.04470 *
## seeded_pool:seeding_time        6.055   3  10.1684 2.512e-06 ***
## seeded_pool:year                3.267  15   1.0972   0.35959
## seeding_time:year              12.315   5  12.4095 1.041e-10 ***
## seeded_pool:seeding_time:year  3.623  15   1.2171   0.25930
## Residuals                      47.237 238
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(m_2)
```

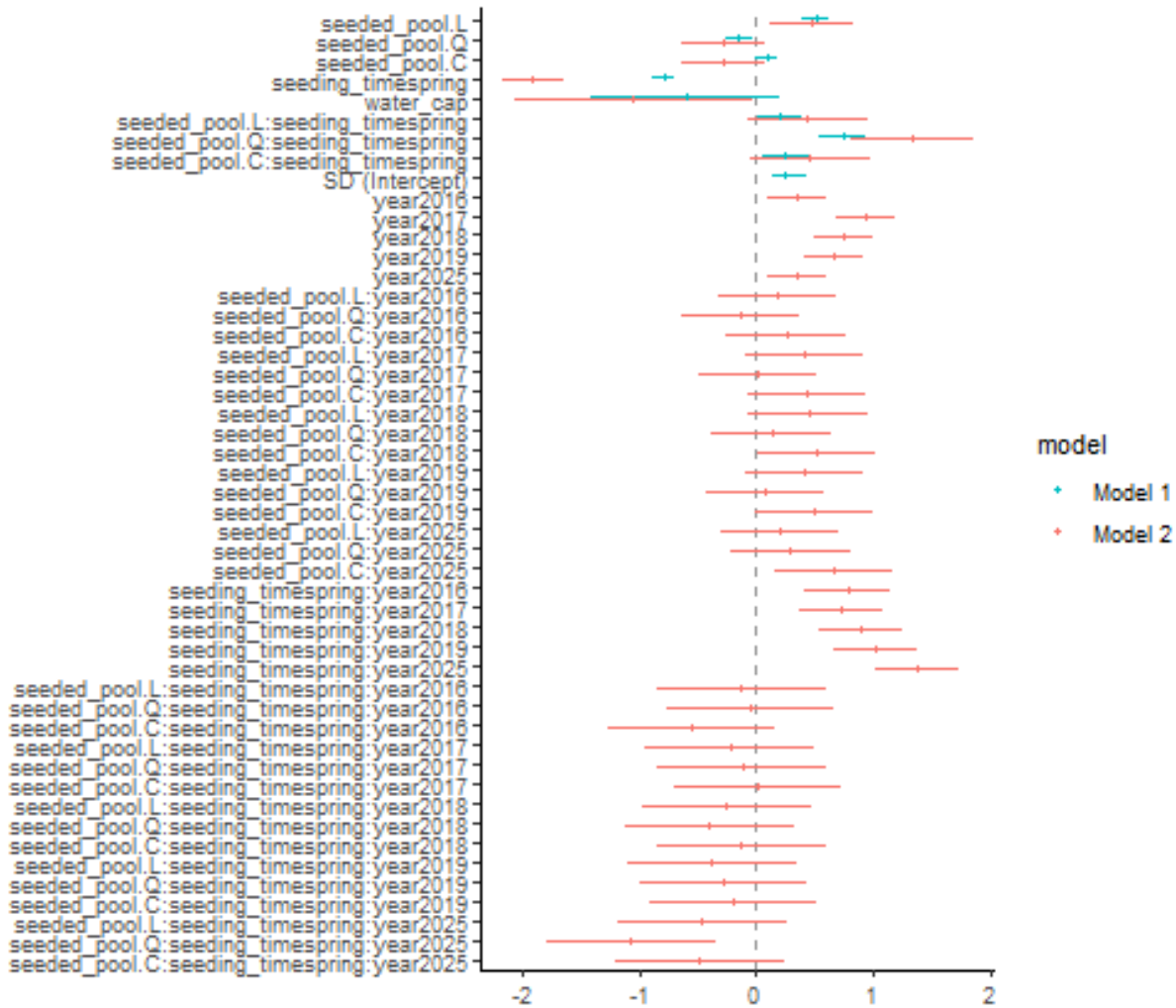
```
##
## Call:
## lm(formula = sqrt(y) ~ seeded_pool * seeding_time * year + water_cap,
##     data = sites)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.7090 -0.2067  0.0238  0.2318  1.3147
##
## Coefficients:
##                                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)                              3.249930   0.284658  11.417  < 2e-16
## seeded_pool.L                            0.474854   0.183812   2.583 0.010382
## seeded_pool.Q                           -0.277760   0.183427  -1.514 0.131281
## seeded_pool.C                           -0.274167   0.182331  -1.504 0.133990
## seeding_timespring                      -1.911590   0.130228 -14.679  < 2e-16
## year2016                                 0.355853   0.128607   2.767 0.006102
## year2017                                 0.928281   0.128607   7.218 7.04e-12
## year2018                                 0.751716   0.128607   5.845 1.66e-08
## year2019                                 0.663779   0.128607   5.161 5.17e-07
## year2025                                 0.351206   0.128607   2.731 0.006790
```

```
## water_cap                                -1.050506   0.520534  -2.018 0.044700
## seeded_pool.L:seeding_timespring          0.441467   0.264390   1.670 0.096283
## seeded_pool.Q:seeding_timespring          1.337209   0.263908   5.067 8.11e-07
## seeded_pool.C:seeding_timespring          0.461191   0.258063   1.787 0.075190
## seeded_pool.L:year2016                    0.183622   0.257214   0.714 0.475995
## seeded_pool.Q:year2016                   -0.142229   0.257214  -0.553 0.580811
## seeded_pool.C:year2016                    0.254713   0.257214   0.990 0.323044
## seeded_pool.L:year2017                    0.418688   0.257214   1.628 0.104894
## seeded_pool.Q:year2017                    0.009668   0.257214   0.038 0.970049
## seeded_pool.C:year2017                    0.435040   0.257214   1.691 0.092078
## seeded_pool.L:year2018                    0.444160   0.257214   1.727 0.085499
## seeded_pool.Q:year2018                    0.128450   0.257214   0.499 0.617965
## seeded_pool.C:year2018                    0.517250   0.257214   2.011 0.045456
## seeded_pool.L:year2019                    0.416378   0.257214   1.619 0.106814
## seeded_pool.Q:year2019                    0.070311   0.257214   0.273 0.784815
## seeded_pool.C:year2019                    0.493517   0.257214   1.919 0.056218
## seeded_pool.L:year2025                    0.198500   0.257214   0.772 0.441039
## seeded_pool.Q:year2025                    0.294737   0.257214   1.146 0.252993
## seeded_pool.C:year2025                    0.671564   0.257214   2.611 0.009603
## seeding_timespring:year2016               0.782501   0.183011   4.276 2.76e-05
## seeding_timespring:year2017               0.723670   0.183011   3.954 0.000101
## seeding_timespring:year2018               0.897979   0.183011   4.907 1.72e-06
## seeding_timespring:year2019               1.024631   0.183011   5.599 5.93e-08
## seeding_timespring:year2025               1.379489   0.183011   7.538 9.98e-13
## seeded_pool.L:seeding_timespring:year2016 -0.132231   0.367827  -0.359 0.719546
## seeded_pool.Q:seeding_timespring:year2016 -0.047153   0.366023  -0.129 0.897604
## seeded_pool.C:seeding_timespring:year2016 -0.548646   0.364210  -1.506 0.133290
## seeded_pool.L:seeding_timespring:year2017 -0.219515   0.367827  -0.597 0.551216
## seeded_pool.Q:seeding_timespring:year2017 -0.120374   0.366023  -0.329 0.742544
## seeded_pool.C:seeding_timespring:year2017  0.006955   0.364210   0.019 0.984779
## seeded_pool.L:seeding_timespring:year2018 -0.259449   0.367827  -0.705 0.481278
## seeded_pool.Q:seeding_timespring:year2018 -0.398030   0.366023  -1.087 0.277940
## seeded_pool.C:seeding_timespring:year2018 -0.126587   0.364210  -0.348 0.728473
## seeded_pool.L:seeding_timespring:year2019 -0.378054   0.367827  -1.028 0.305086
## seeded_pool.Q:seeding_timespring:year2019 -0.278182   0.366023  -0.760 0.447999
## seeded_pool.C:seeding_timespring:year2019 -0.198583   0.364210  -0.545 0.586097
## seeded_pool.L:seeding_timespring:year2025 -0.465371   0.367827  -1.265 0.207040
## seeded_pool.Q:seeding_timespring:year2025 -1.075427   0.366023  -2.938 0.003626
## seeded_pool.C:seeding_timespring:year2025 -0.482489   0.364210  -1.325 0.186522
##
## (Intercept)                               ***
## seeded_pool.L                             *
## seeded_pool.Q
## seeded_pool.C
## seeding_timespring                        ***
## year2016                                  **
## year2017                                  ***
## year2018                                  ***
## year2019                                  ***
## year2025                                  **
## water_cap                                 *
## seeded_pool.L:seeding_timespring          .
## seeded_pool.Q:seeding_timespring          ***
## seeded_pool.C:seeding_timespring          .
```

```
## seeded_pool.L:year2016
## seeded_pool.Q:year2016
## seeded_pool.C:year2016
## seeded_pool.L:year2017
## seeded_pool.Q:year2017
## seeded_pool.C:year2017                    .
## seeded_pool.L:year2018                    .
## seeded_pool.Q:year2018
## seeded_pool.C:year2018                    *
## seeded_pool.L:year2019
## seeded_pool.Q:year2019
## seeded_pool.C:year2019                    .
## seeded_pool.L:year2025
## seeded_pool.Q:year2025
## seeded_pool.C:year2025                    **
## seeding_timespring:year2016               ***
## seeding_timespring:year2017               ***
## seeding_timespring:year2018               ***
## seeding_timespring:year2019               ***
## seeding_timespring:year2025               ***
## seeded_pool.L:seeding_timespring:year2016
## seeded_pool.Q:seeding_timespring:year2016
## seeded_pool.C:seeding_timespring:year2016
## seeded_pool.L:seeding_timespring:year2017
## seeded_pool.Q:seeding_timespring:year2017
## seeded_pool.C:seeding_timespring:year2017
## seeded_pool.L:seeding_timespring:year2018
## seeded_pool.Q:seeding_timespring:year2018
## seeded_pool.C:seeding_timespring:year2018
## seeded_pool.L:seeding_timespring:year2019
## seeded_pool.Q:seeding_timespring:year2019
## seeded_pool.C:seeding_timespring:year2019
## seeded_pool.L:seeding_timespring:year2025
## seeded_pool.Q:seeding_timespring:year2025 **
## seeded_pool.C:seeding_timespring:year2025
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4455 on 238 degrees of freedom
## Multiple R-squared:  0.8404, Adjusted R-squared:  0.8082
## F-statistic: 26.11 on 48 and 238 DF,  p-value: < 2.2e-16
```

Forest plot

```r
dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  theme_classic()
```



Effect sizes

Effect sizes of chosen model just to get exact values of means etc. if necessary.

```
(emm <- emmeans(
  m_2,
  revpairwise ~ seeding_time | seeded_pool | year,
  type = "response"
  ))
```

```
## $emmeans
## seeded_pool = 6, year = 2015:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall           5.3466 0.8420 238   3.8169    7.134
##  spring         0.4489 0.2440 238   0.0968    1.059
##
## seeded_pool = 12, year = 2015:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall           6.5391 0.9310 238   4.8336    8.502
##  spring         0.0352 0.0684 238   0.0000    0.299
```

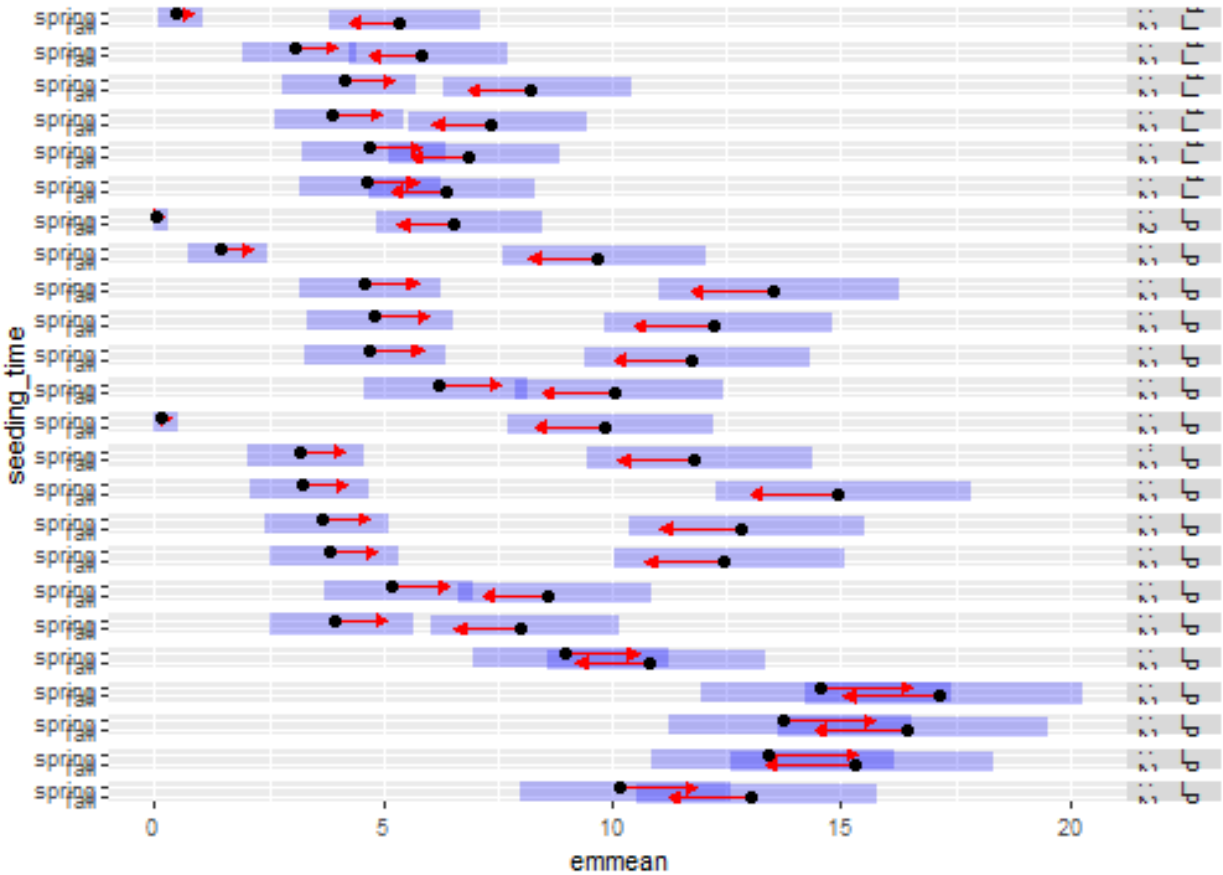
```
##
## seeded_pool = 18, year = 2015:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall           9.8431 1.1400 238   7.7180   12.226
##  spring         0.1201 0.1260 238   0.0000    0.497
##
## seeded_pool = 33, year = 2015:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall           7.9905 1.0500 238   6.0626   10.184
##  spring         3.9324 0.7910 238   2.5280    5.646
##
## seeded_pool = 6, year = 2016:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall           5.8413 0.8800 238   4.2366    7.703
##  spring         3.0448 0.6360 238   1.9211    4.426
##
## seeded_pool = 12, year = 2016:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall           9.6967 1.1300 238   7.5918   12.059
##  spring         1.4690 0.4420 238   0.7278    2.468
##
## seeded_pool = 18, year = 2016:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall          11.7960 1.2500 238   9.4575   14.393
##  spring         3.1978 0.6510 238   2.0441    4.609
##
## seeded_pool = 33, year = 2016:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall          10.8348 1.2200 238   8.5680   13.367
##  spring         8.9727 1.0900 238   6.9499   11.253
##
## seeded_pool = 6, year = 2017:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall           8.2211 1.0400 238   6.2934   10.406
##  spring         4.1379 0.7410 238   2.8066    5.727
##
## seeded_pool = 12, year = 2017:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall          13.5338 1.3400 238  11.0238   16.301
##  spring         4.6092 0.7820 238   3.1971    6.279
##
## seeded_pool = 18, year = 2017:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall          14.9197 1.4100 238  12.2736   17.824
##  spring         3.2467 0.6560 238   2.0832    4.667
##
## seeded_pool = 33, year = 2017:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall          17.1231 1.5300 238  14.2393   20.273
##  spring        14.5319 1.3900 238  11.9225   17.399
##
## seeded_pool = 6, year = 2018:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall           7.3691 0.9880 238   5.5508    9.445
```

```
##  spring         3.8953 0.7190 238   2.6075    5.441
##
## seeded_pool = 12, year = 2018:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall          12.1964 1.2700 238   9.8201   14.830
##  spring         4.8088 0.7990 238   3.3636    6.512
##
## seeded_pool = 18, year = 2018:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall          12.7963 1.3000 238  10.3553   15.495
##  spring         3.6490 0.6950 238   2.4078    5.147
##
## seeded_pool = 33, year = 2018:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall          16.4536 1.5000 238  13.6293   19.544
##  spring        13.7582 1.3500 238  11.2227   16.552
##
## seeded_pool = 6, year = 2019:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall           6.8725 0.9540 238   5.1209    8.881
##  spring         4.6779 0.7880 238   3.2543    6.359
##
## seeded_pool = 12, year = 2019:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall          11.7221 1.2500 238   9.3950   14.306
##  spring         4.7057 0.7900 238   3.2775    6.392
##
## seeded_pool = 18, year = 2019:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall          12.4471 1.2900 238  10.0413   15.111
##  spring         3.8009 0.7100 238   2.5315    5.327
##
## seeded_pool = 33, year = 2019:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall          15.3300 1.4500 238  12.6085   18.317
##  spring        13.3895 1.3300 238  10.8899   16.147
##
## seeded_pool = 6, year = 2025:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall           6.3884 0.9200 238   4.7043    8.330
##  spring         4.6101 0.7820 238   3.1978    6.280
##
## seeded_pool = 12, year = 2025:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall          10.0307 1.1500 238   7.8877   12.431
##  spring         6.2260 0.9090 238   4.5638    8.146
##
## seeded_pool = 18, year = 2025:
##  seeding_time response     SE  df lower.CL upper.CL
##  fall           8.6148 1.0700 238   6.6351   10.853
##  spring         5.2032 0.8300 238   3.6961    6.967
##
## seeded_pool = 33, year = 2025:
##  seeding_time response     SE  df lower.CL upper.CL
```

```
##  fall          13.0223 1.3400 238  10.5244   15.786
##  spring        10.1547 1.1600 238   7.9945   12.573
##
## Confidence level used: 0.95
## Intervals are back-transformed from the sqrt scale
##
## $contrasts
## seeded_pool = 6, year = 2015:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -1.642 0.257 238  -6.384  <.0001
##
## seeded_pool = 12, year = 2015:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -2.370 0.258 238  -9.189  <.0001
##
## seeded_pool = 18, year = 2015:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -2.791 0.258 238 -10.819  <.0001
##
## seeded_pool = 33, year = 2015:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -0.844 0.273 238  -3.085  0.0023
##
## seeded_pool = 6, year = 2016:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -0.672 0.257 238  -2.612  0.0096
##
## seeded_pool = 12, year = 2016:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -1.902 0.258 238  -7.376  <.0001
##
## seeded_pool = 18, year = 2016:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -1.646 0.258 238  -6.382  <.0001
##
## seeded_pool = 33, year = 2016:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -0.296 0.261 238  -1.133  0.2584
##
## seeded_pool = 6, year = 2017:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -0.833 0.257 238  -3.239  0.0014
##
## seeded_pool = 12, year = 2017:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -1.532 0.258 238  -5.941  <.0001
##
## seeded_pool = 18, year = 2017:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -2.061 0.258 238  -7.988  <.0001
##
## seeded_pool = 33, year = 2017:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -0.326 0.261 238  -1.247  0.2137
```

```
##
## seeded_pool = 6, year = 2018:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -0.741 0.257 238  -2.880  0.0043
##
## seeded_pool = 12, year = 2018:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -1.299 0.258 238  -5.039  <.0001
##
## seeded_pool = 18, year = 2018:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -1.667 0.258 238  -6.462  <.0001
##
## seeded_pool = 33, year = 2018:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -0.347 0.261 238  -1.328  0.1856
##
## seeded_pool = 6, year = 2019:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -0.459 0.257 238  -1.783  0.0758
##
## seeded_pool = 12, year = 2019:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -1.254 0.258 238  -4.865  <.0001
##
## seeded_pool = 18, year = 2019:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -1.578 0.258 238  -6.119  <.0001
##
## seeded_pool = 33, year = 2019:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -0.256 0.261 238  -0.980  0.3281
##
## seeded_pool = 6, year = 2025:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -0.380 0.257 238  -1.479  0.1405
##
## seeded_pool = 12, year = 2025:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -0.672 0.258 238  -2.606  0.0097
##
## seeded_pool = 18, year = 2025:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -0.654 0.258 238  -2.535  0.0119
##
## seeded_pool = 33, year = 2025:
##  contrast      estimate    SE  df t.ratio p.value
##  spring - fall   -0.422 0.261 238  -1.614  0.1078
##
## Note: contrasts are still on the sqrt scale. Consider using
##       regrid() if you want contrasts of back-transformed estimates.
```

```r
plot(emm, comparison = TRUE)
```



Session info

```
## R version 4.5.0 (2025-04-11 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_1.11.1   DHARMa_0.4.7     glmmTMB_1.1.12   lme4_1.1-37
##  [5] Matrix_1.7-3     patchwork_1.3.1  ggbeeswarm_0.7.2 lubridate_1.9.4
##  [9] forcats_1.0.0    stringr_1.5.1    dplyr_1.1.4      purrr_1.1.0
```

```
## [13] readr_2.1.5      tidyr_1.3.1      tibble_3.3.0     ggplot2_3.5.2
## [17] tidyverse_2.0.0  here_1.0.1
##
## loaded via a namespace (and not attached):
##  [1] Rdpack_2.6.4           gridExtra_2.3          sandwich_3.1-1
##  [4] rlang_1.1.6            magrittr_2.0.3         compiler_4.5.0
##  [7] mgcv_1.9-3             vctrs_0.6.5            pkgconfig_2.0.3
## [10] crayon_1.5.3           fastmap_1.2.0          backports_1.5.0
## [13] labeling_0.4.3         utf8_1.2.6             ggstance_0.3.7
## [16] promises_1.3.3         rmarkdown_2.29         tzdb_0.5.0
## [19] nloptr_2.2.1           bit_4.6.0              xfun_0.52
## [22] later_1.4.2            parallel_4.5.0         R6_2.6.1
## [25] gap.datasets_0.0.6     stringi_1.8.7          qgam_2.0.0
## [28] RColorBrewer_1.1-3     car_3.1-3              boot_1.3-31
## [31] numDeriv_2016.8-1.1    estimability_1.5.1     Rcpp_1.1.0
## [34] iterators_1.0.14       knitr_1.50             zoo_1.8-14
## [37] parameters_0.27.0      httpuv_1.6.16          splines_4.5.0
## [40] timechange_0.3.0       tidyselect_1.2.1       rstudioapi_0.17.1
## [43] abind_1.4-8            yaml_2.3.10            MuMIn_1.48.11
## [46] doParallel_1.0.17      TMB_1.9.17             codetools_0.2-20
## [49] lattice_0.22-7         plyr_1.8.9             shiny_1.11.1
## [52] withr_3.0.2            bayestestR_0.16.1      coda_0.19-4.1
## [55] evaluate_1.0.4         marginaleffects_0.28.0 pillar_1.11.0
## [58] gap_1.6                carData_3.0-5          foreach_1.5.2
## [61] stats4_4.5.0           reformulas_0.4.1       insight_1.3.1
## [64] generics_0.1.4         vroom_1.6.5            rprojroot_2.0.4
## [67] hms_1.1.3              scales_1.4.0           minqa_1.2.8
## [70] xtable_1.8-4           glue_1.8.0             tools_4.5.0
## [73] data.table_1.17.8      mvtnorm_1.3-3          grid_4.5.0
## [76] rbibutils_2.3          datawizard_1.1.0       nlme_3.1-168
## [79] Rmisc_1.5.1            performance_0.15.0     beeswarm_0.4.0
## [82] vipor_0.4.7            Formula_1.2-5          cli_3.6.5
## [85] gtable_0.3.6           digest_0.6.37          farver_2.1.2
## [88] htmltools_0.5.8.1      lifecycle_1.0.4        mime_0.13
## [91] bit64_4.6.0-1          dotwhisker_0.8.4       MASS_7.3-65
```